

Week 3: Fundamentals of Probability and Statistics I

PLSC 309 Quantitative Political Analysis

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Office Hours Schedule Change

- **Mondays, 1:00–2:30 p.m.**, starting from next week.

Final Project: Checkpoint 1

Due: next Wednesday, September 16 at 11:59 p.m. (midnight)

Instructions: Posted on Canvas

Question 6

Countries with higher levels of trade openness are more likely to join international organizations.

- IV: Level of trade openness/Trade openness
- DV: Joining IOs/IO membership/Likelihood of joining IOs
- Direction: Higher IV $\rightarrow$ greater likelihood of DV

A hypothesis usually contains an implicit or explicit comparison.

Remember: The IV and DV are the concepts that vary, not their particular values or the predicted direction

Feedback: Writing a Hypothesis

A hypothesis is usually **a single sentence** stating the predicted association or causal relationship between an independent variable (IV) and a dependent variable (DV).

Some standard examples:

- [Cases with a higher/lower level of X] are more/less likely to experience Y.
- As X increases, Y increases/decreases.
- X has a positive/negative effect on Y.
- [Cases with X] are more/less likely to experience Y than [cases without X].
- [Change (increase or decrease) in X] increases/decreases Y.

...

What Did We Learn Last Time?

- Random variable
- Probability distribution
- Probability distribution functions (PMF, PDF, CDF)
- Expected value
- Variance and standard deviation
- Covariance and correlation

Review: Random Variable

Random Variable

A **random variable** is a **function** $X : \Omega \rightarrow \mathbb{R}$ that assigns a real number to each outcome in the sample space.

- The *input* is an outcome $\omega \in \Omega$ (which outcome occurs)
- The *output* is a number $X(\omega) \in \mathbb{R}$ (the numerical value)

The “**randomness**” comes from not knowing which ω will occur. (Think about the pollster example)

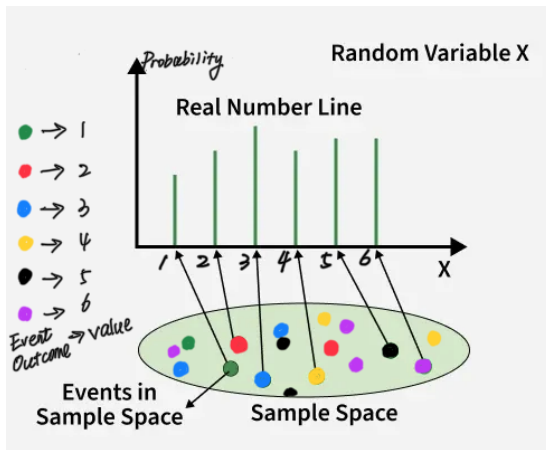
Non-random variable (fixed, deterministic, constant):

- e.g., π (pi), the speed of light in a vacuum (a universal physical constant),...

Review: Probability Distribution

Probability Distribution

A **probability distribution** describes how the possible values of a random variable are distributed along with their chances of occurring.



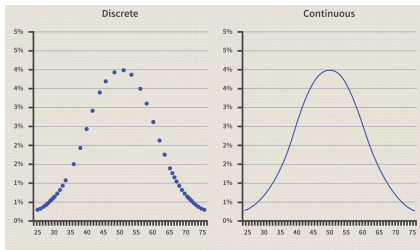
Review: Probability Mass Function vs. Probability Density Function

PMF

For a discrete random variable X :

$$p_X(x) = P(X = x)$$

The exact probability for each value

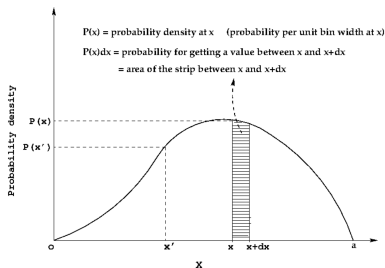


PDF

For a continuous random variable X :

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

The area under the curve between two specific points



Review: Expected Value

A **single number** used to summarize **the center** of a probability distribution.

Expected Value

The **expected value** (or **mean**) of a random variable X is:

$$\mathbb{E}[X] = \begin{cases} \sum x \cdot p(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x \cdot f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$

Interpretation: The expected value is a **weighted average** of possible values, where the weights are the probabilities (or densities).

Notation: $\mathbb{E}[X]$, $\bar{X}$, μ , and μ_X all denote the same thing.

Review: Variance

A **single number** used to describe **the spread** of possible values around the expected value.

Variance

The **variance** of a random variable X is:

$$\text{Var}(X) = \mathbb{E} [(X - \mathbb{E}[X])^2]$$

- Deviations above and below the mean ($X - \mathbb{E}[X]$) would **cancel out**
- Squaring makes all deviations **non-negative**
- Larger deviations get **penalized more heavily**
- A single number: take the average of the **squared deviation from the mean**.

Review: Standard Deviation

Variance has a problem for interpretation because it uses **squared units** given that its calculation involves squaring the differences between each data point and the mean.

Standard deviation: **A single number** used to describe **the spread** of possible values around the expected value in its original unit.

Standard Deviation

The **standard deviation** of X is the square root of its variance:

$$\sigma_X = \sqrt{\text{Var}(X)}$$

Review: Covariance

A **single number** used to quantify how two variables vary together.

Covariance

The **covariance** between two random variables X and Y is:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

It is the expected value of the product of their deviations from their respective means.

$X - \mathbb{E}[X]$	$Y - \mathbb{E}[Y]$	Product
+	+	+
-	-	+
+	-	-
-	+	-

Review: Correlation

A problem with covariance is that **its magnitude depends on the units of measurement**.

Correlation: **a single number** between -1 and 1 that shows **the strength and direction** of a relationship between two variables

Correlation

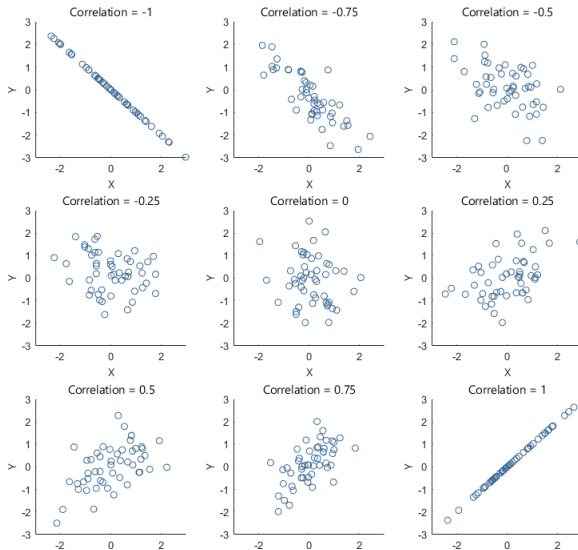
The **correlation** between X and Y is:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Correlation	Interpretation
$\rho = 1$	Perfect positive linear relationship
$\rho \approx 0$	Little or no linear relationship
$\rho = -1$	Perfect negative linear relationship

Correlation Coefficient

Realizations of couples of random variables X and Y
with different correlation coefficients



Today's Agenda

- Sampling and Estimation (learning population from samples)

Acknowledgement: Some slides in today's lecture are adapted from Scott Cunningham's [GOV 2001 course materials at Harvard](#) and from Brandon Stewart's [SOC 500 course materials at Princeton](#).

Sampling and Estimation

The Practical Problem

You are an analyst for a candidate running for office in Pennsylvania.

Your boss asks you a simple question:

What percentage of Pennsylvania voters support our candidate?

Why does this matter?

- The campaign wants to assess its **chance of winning**.
- Your boss wants a **single number**:

Support = ?%

But there is a practical problem:

You cannot ask every voter in Pennsylvania.

Access, time, and cost make surveying the entire voting population practically impossible.

So What Can We Do?

We want to know:

Support among **all** Pennsylvania voters

But we can only observe:

Support among **some** Pennsylvania voters

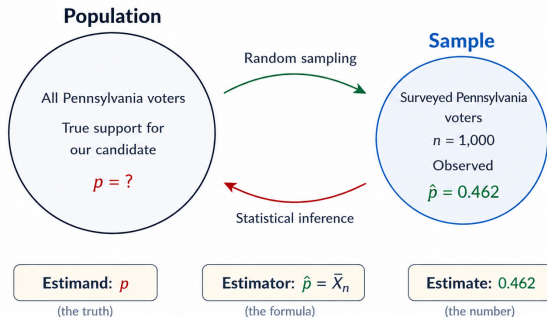
So we:

- 1 Select a **random sample** of voters.
- 2 Measure support in that random sample.
- 3 Use the sample to **estimate** support in the population.

Population → **Sample** → **Estimate**

Learning from Random Samples

From Population to Sample



Statistical inference is the process of using data from a (small) random sample to draw conclusions, test claims, or estimate unknown properties of a (large) population.

From Finite Population to Random Sample

A finite population of N units with values $x_1, \dots, x_N$.

Population mean:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

Population variance:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Random sampling: Draw n units at random. Each draw X_i has the same distribution (**identically distributed**) and draws do not affect each other (**independent**).

I.I.D.: $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$

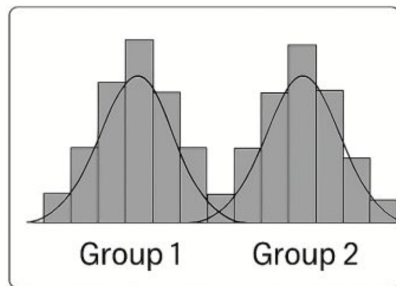
❶ **Independent:**

$$X_i \perp X_j \quad \text{for all } i \neq j$$

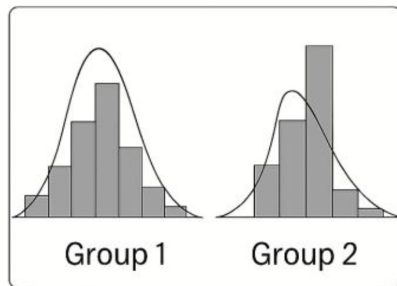
❷ **Identically distributed:** Each X_i has the same probability distribution.

Note: x_i denotes the fixed value of unit i in the population, whereas X_i denotes the random value obtained on the i -th draw from the population. Thus, x_i is fixed, while X_i is a random variable.

Assumptions met



Different shapes



IID: Independent and Identically Distributed —

Formalizing the Framework

Precise definitions for the visual we just saw

Estimand $\theta = T(F)$:

- The specific target quantity or population parameter that you want to calculate or estimate in a statistical study.
- Examples: $\mu = \mathbb{E}[X]$, $\sigma^2 = \text{Var}(X)$

Estimator $\hat{\theta}_n = h(X_1, \dots, X_n)$:

- A function of the sample—a **random variable** (it has a distribution!)

Estimate:

- A realized value of the estimator—a *number*, not a random variable

Common mistake: “My estimator was 0.462.” No—your *estimate* was 0.462. The estimator is the formula.

Estimand, Estimator, and Estimate (in plain language)

In our Pennsylvania polling example:

p
Estimand

$\hat{p} = \bar{X}_n$
Estimator

0.462
Estimate

Estimand

The population quantity
we want to know.

p

Fixed but unknown

Estimator

The rule we use to
estimate the estimand
from a sample.

$\hat{p} = \bar{X}_n$

Random variable

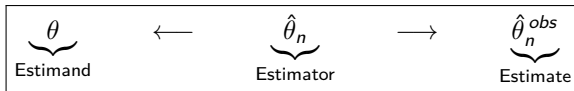
Estimate

The value produced by
the estimator in this
sample.

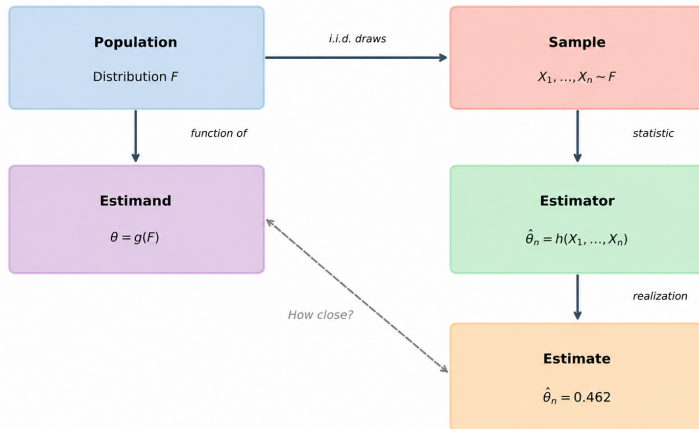
0.462

A realized number

General notation:



The Estimation Framework



The Sample Mean as an Estimator

Natural idea: Estimate μ with the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Key insight: $\bar{X}_n$ is itself a *random variable*.

- Different samples give different values of $\bar{X}_n$
- $\bar{X}_n$ has its own distribution—the **sampling distribution**
- We want to understand this distribution

In our Pennsylvania polling example: $\bar{X}_n = 0.462$ is one draw from the sampling distribution of the sample mean.

Statistics is about understanding how estimators behave across repeated samples.

The Three Distributions: What Are They?

① **Population Distribution** $F(x)$:

The true distribution of values in the population—aka. the **data-generating process (DGP)**.

② **Empirical Distribution**:

The distribution we observe in the sample $X_1, \dots, X_n$ —what we actually have. For our polling example, this is a series of 1s and 0s.

③ **Sampling Distribution** $G(\hat{\theta}_n)$:

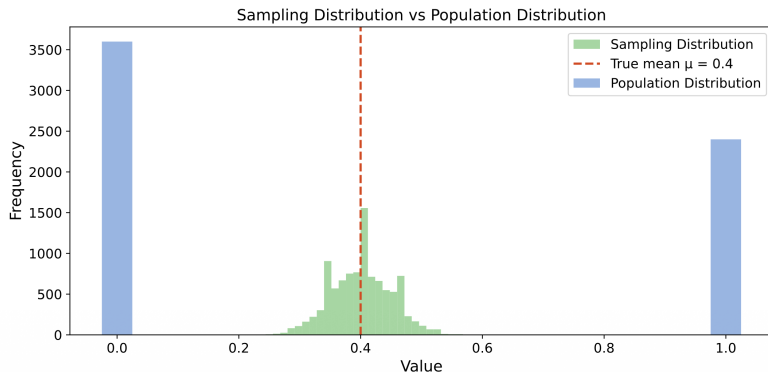
The distribution of the **estimator** over repeated samples.

Our observed sample mean, $\bar{X}_n = 0.462$, is *one draw* from this distribution.

Trick question: “The sampling distribution is the distribution of θ .” True or false?

False—it is the distribution of $\hat{\theta}_n$, the estimator.

Sampling Distribution vs. Population Distribution



Population is binary (0s and 1s). But the sampling distribution of $\bar{X}_n$ is bell-shaped and concentrated around the true mean.

Simulating One Poll in R

Suppose the true support for our candidate is:

$$p = 0.40$$

and we survey $n = 1,000$ Pennsylvania voters.

```
set.seed(123)
p_true <- 0.40 # true parameter (the estimand)
n <- 1000 # sample size

# Simulate n=1000 voters: 1 = support, 0 = not support
support <- rbinom(n, size = 1, prob = p_true)
# Look at the first six simulated voters
head(support)
```

```
## [1] 0 1 0 1 1 0
```

```
# Calculate the sample mean
mean(support)
```

```
## [1] 0.393
```

What If We Repeat the Poll?

Now suppose we conduct the same poll 1000 times.

```
R <- 1000 # 1000 draws (random sampling)
```

```
estimates <- replicate(  
  R,  
  mean(rbinom(n, size = 1, prob = p_true))  
)
```

```
head(estimates) # the first six sample means
```

```
## [1] 0.386 0.417 0.409 0.384 0.396 0.390
```

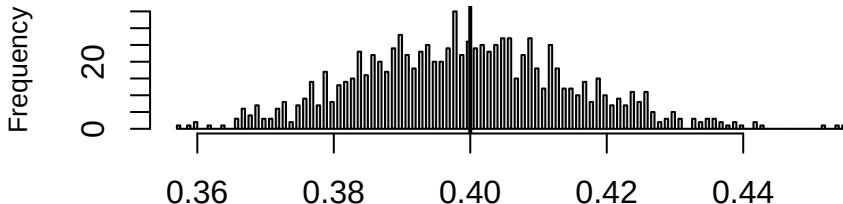
```
mean(estimates) # expected value of the sample mean
```

```
## [1] 0.399405
```

The Sampling Distribution in R

```
hist(  
  estimates,  
  breaks = 150,  
  main = "Sampling Distribution of the Sample Mean",  
  xlab = "Estimated Support",  
  cex.main = 0.8, cex.lab = 0.8, # customize label sizes  
)  
  
abline(v = p_true, lwd = 2)
```

Sampling Distribution of the Sample Mean



Key Takeaways

- Statistical inference
- Random sampling (iid.)
- Estimand, estimator, estimate
- Sampling distribution